The Value of Fleet-Resource Coupling: Objective Levels and Physical Realization

A research section accompanying the stable GI-TBC method

1 Inventory proof and the physical optimum

The stable Gini-interval tailored branch-and-cut method (GI-TBC) maintains complete improving-domain coverage, a ledger of scope-valid bounds, and independently verified fleet incumbents. Integer-inventory pricing can strengthen the ledger without changing the interval controller or route producer. This separation makes it possible to study a stronger relaxation without silently making it part of the complete algorithm. The default method remains GI-TBC/B; E+ is the enhanced comparison baseline inherited from the original-objective inventory study.

Let $r_i = c_i Y_i$, $S = \sum_i r_i$, and

$$G(Y) = \frac{\sum_{i<j} |r_i - r_j|}{nS}, \qquad H(Y) = \sum_i \omega_i |r_i - 1|, \qquad J(Y) = G(Y) + H(Y).$$

The original zero convention is retained. The coefficient ω_i already includes λ. All domains below lie in the same qualified improving inventory scope, with the original supports, hard total and coverage rules. The present scopes have $\gamma_L = 0$ and a strictly positive stock denominator.

Write D_I for the integer inventory relaxation. For RX35, let D_F delete the 26 vectors excluded by the two preceding RX35 family certificates. This deletion set is specific to RX35; for the NB35 comparison take $D_F = D_I$. Later family exclusions do not silently redefine D_F. Let $D_R(T)$ be the inventory projection of free vehicle assignment, per-vehicle pickup/drop and return-load balance, handling, and valid closed-tour lower bounds for at most three stations. Finally, $D_P(T)$ contains inventories realized by a complete original fleet. Under the inherited physical and numerical qualification,

$$D_P(T) \subseteq D_R(T) \subseteq D_F \subseteq D_I, \qquad F_I \le F_F \le F_R \le F_P,$$

where $F_D = \min_{Y \in D} J(Y)$. A strict set inclusion does not by itself make the corresponding minimum strictly larger. In particular, the best saved inventory families are not a proved ordering of the global k-best inventories.

Proposition 1 (A resource witness bounds same-source cut value)**.** *Suppose $L_I \le F_I$ and a fully checked original-T resource assignment realizes inventory Y^R of price $U_R = J(Y^R)$. If a cut domain D_C satisfies $D_R(T) \subseteq D_C \subseteq D_I$, then*

$$F_C \le F_R \le U_R, \qquad 0 \le F_R - F_I \le U_R - L_I.$$

Proof. Domain containment orders the minima; Y^R supplies an actual member of the resource domain. The inventory lower bounds F_I from below. □

Thus a low resource witness limits every original-objective floor derived solely from rearranging the same resource facts, including projection, core and no-good cuts. The limit does not apply to stronger travel or prefix-load information, or to producing a real route. An inventory-only upper cannot play the role of U_R. A known physical route supplies a valid but potentially loose resource upper by projection; a resource assignment alone supplies no physical upper.

2 Whole objective levels, rather than isolated exclusions

The principal question can be posed as the hard level

$$\mathcal{R}(\tau, T) = \{(Y, a) : Y \in D_I,\ J(Y) \le \tau,\ a \text{ satisfies the full necessary resource model}\}.$$

A qualified infeasibility result excludes the complete level, so $F_R > \tau$ and $F_P > \tau$. Within the inherited proof policy, a conservative recorded lower is τ. Feasibility instead requires a complete saved integer inventory and assignment, followed by independent evaluation of the true objective and original time conditions. An unfinished query does not improve either side of the prior interval.

The predeclared primary level is

$$\tau_* = L_I + \tfrac{1}{4}(U_P - L_I).$$

For RX35 this is approximately 0.2519564273646922. Excluding it would remove one quarter of the inherited physical gap. Since the known inventory upper is approximately 0.23612171136420015, that conclusion would also establish a structural resource/inventory difference of about 0.015834716. These are implications of a complete-level proof, not consequences of excluding further stored points.

The normalized inventory formulation uses a Gini epigraph q. At an integer stock with $\gamma_L = 0$, $q = G(Y)$ is available; hence $q + H \le \tau$ covers every true low-objective stock. For each station choose an exact coefficient anchor a_i and a power-of-two scale s. Actual emitted coefficients $\widehat{c}_{iy}$ represent $(h_i(y) - a_i)/s$, rounded downward except for deliberate tiny omissions. Let

$$\epsilon = \sum_i \max\bigl(0, \max_y\{\widehat{c}_{iy} - (h_i(y) - a_i)/s\}\bigr).$$

The actual cutoff is

$$q/s + \sum_{i,y} \widehat{c}_{iy} x_{iy} \le \operatorname{up}\left((\tau - \sum_i a_i)/s + \epsilon\right).$$

Every coefficient and the RHS are checked in the Native matrix. The one-hot relations bound the total positive coefficient excess, so no omitted tiny coefficient silently deletes a valid low-objective point. A returned point is still repriced exactly; a relaxed cutoff value is not sufficient to assert membership in the true level. The old known-lower auxiliary row is not used in these level models.

3 A qualified integer budget expression

The resource projection inherits classical vehicle-resource and tour necessary conditions. Its integer rounding follows the earlier rounded handling and anchor/whole-handling studies; it is not a new general cut theory. The old production H-ACT mechanism remains parked.

Let $h > 0$ be a common lower handling coefficient and $\ell(S)$ a valid closed-tour lower for visiting S. Raw physical and emitted canonical coefficients first share a coefficient-wise lower envelope. The shortest-path and subset-tour computations use exact binary-rational arithmetic and downward

enclosures. A common upper horizon $\overline{T}$ covers the original physical acceptance rule, its arithmetic envelope, and the inherited time guard. Both tested expressions use the identical coefficient package; a bare floor at nominal T is not substituted.

Total pickups P_k are integral sums of integer quantities and binary service decisions. Define

$$U = \left\lfloor \frac{\overline{T}}{h} \right\rfloor, \qquad b_S = \left\lfloor \frac{\overline{T} - \ell(S)}{h} \right\rfloor.$$

Always $P_k \le U$. For $0 \le b_S < U$, replace the real-valued time row by

$$P_k + (U - b_S) \sum_{i \in S} s_{ki} \le b_S + (U - b_S)|S|. \tag{1}$$

If $b_S < 0$, impose $\sum_{i \in S} s_{ki} \le |S| - 1$. Total pickups are never incorrectly bounded by Q_k: deliveries can free space for later pickups, while the physical capacity applies to prefix loads and the returned load.

Proposition 2 (Integer equivalence and subset dominance)**.** *In the same coefficient package, (1), the forbidden-set rows and $P_k \le U$ preserve the integer service domain and imply the old real-valued resource rows in the LP relaxation. If a proper subset $R \subset S$ has $b_R \le b_S$, its budget row and the global budget imply the S row. A forbidden subset forbids its supersets.*

Proof. Put $\delta_S = |S| - \sum_{i \in S} s_{ki}$. At integer visits, $\delta_S = 0$ activates the exact integer budget, whereas $\delta_S \ge 1$ leaves the global budget sufficient. For $0 \le \delta_S \le 1$, the new bound connects the downward-rounded budgets at both endpoints, and therefore lies below the old real-valued time bound. For $\delta_S \ge 1$, $P_k \le U$ already implies that old bound. For dominance, $\delta_R \le \delta_S$ and, on $[0, 1]$, $b_R + (U - b_R)\delta_S \le b_S + (U - b_S)\delta_S$; the global budget covers the remaining range. The forbidden-set statement follows directly. □

Under the qualified acceptance horizon, RX35 retains 4087 of 7175 subset rows per vehicle; its 264 negative budgets reduce to 81 minimal forbidden sets. NB35 retains 4755 subsets, with four minimal forbidden sets. Two RX35 and five NB35 budgets cross an integer threshold when the time acceptance margin is included. These counts explain the scale of the expression opportunity; they establish no speedup on their own.

3.1 A fractional ownership obstruction

An integer stock can be difficult for the resource model even when its inventory variables already have an exact normalized lift. The following elementary construction separates inventory integrality from vehicle ownership integrality. It concerns the explicit bare LPs, before additional integer-valid cuts or branching.

Proposition 3. *Let an integer stock have an inventory LP lift in the hard level. Suppose there are $M \ge 2$ vehicles, every nonzero service is individually eligible for every vehicle, and its total pickups/drops P, D satisfy*

$$0 \le (P - D)/M \le \min_k Q_k, \qquad P \le (M - 1)U.$$

If every singleton tour lower is at most $\overline{T}$, assigning a fraction $1/M$ of each nonzero service to every vehicle extends that stock to a feasible LP point of both ideal resource expressions, including all their rounded and forbidden-subset rows.

Proof. Each vehicle receives pickup P/M, drop D/M and visit value at most $1/M$ per station. Assignment and return balance hold. The global budget holds since $P/M \le (M-1)U/M \le U$. For a singleton, the old time row is at most $(hP+\overline{T})/M \le \overline{T}$. For $|S| \ge 2$, $\sum_{i\in S} s_{ki} - |S| + 1 \le 0$, so handling alone suffices for the old row. For a rounded nonforbidden singleton, $P/M \le (M-1)U/M \le b_S + (U-b_S)(1-1/M)$. For larger sets, $\delta_S \ge 1$ makes the global budget sufficient. A forbidden set has size at least two under the singleton assumption, and $\sum_{i\in S} s_{ki} \le |S|/M \le |S|-1$. □

This is a standard fractional-assignment weakness, not an integer resource upper. A checked instantiation at price $J(Y)$ can upper-bound the original-price optimum of those fixed bare LPs. It cannot upper-bound F_R, and additional integer-valid cuts can remove it. In particular, survival of one impossible stock in the LP does not establish a strict gap between the global inventory and resource minima. Actual binary-matrix coefficients and RHS values require a separate saved-point check before an exact numerical LP witness is reported.

4 Finite queries and a complete hard-level campaign

Two preselected free-assignment queries were tested first. RX35's retained Y3 exact-price family contains ten members; NB35's saved-best family is a singleton. Both were excluded at their original/guarded time horizon. The queried v1 distance reconstruction gives necessary-time lowers 6310.129749041621 and 7206.982290055626 seconds, respectively, against original horizons 6300 and 7200. These numeric time lowers belong to the queried v1 models. A finite audit of the actual C++ physical duration rows found identical integer budgets at both nominal and guarded horizons, which transfers the fixed-time exclusions; it does not relabel the old numbers as identical unconditional C++-coefficient time minima. Their relaxed-time assignments are neither original-time resource uppers nor physical routes. Family scope remains explicit.

Role	L_I	U_I	Inherited $U_R = U_P$
RX35	.2360521029	.2361217114	.2996694007
NB35	.3959564087	.3960465490	.4502915033

Table 1: Retained coarse root-scope intervals. The physical upper projects to a resource upper; no novel low-price integer resource assignment or new physical upper was obtained. RX35's D_F upper remains the inventory-only Y3 price, approximately .2368540093.

The one qualified expression alternative retains 4087 RX35 subsets instead of 7175, reducing the full model from 24154 to 14890 rows while keeping 2673 columns. A finite raw diagnostic returned at 594 nodes in 753.307 outer seconds; the rounded diagnostic reached 512 nodes in 847.824 seconds. These are unequal observed node counts, not a matched-node speed comparison. No compression speedup was demonstrated, and the raw expression was selected for the substantive capacity.

The actual C++-coefficient raw hard level at $\tau_* = .2519564273646922$ then completed its full declared 129600-second Native window. All inventory and vehicle decisions remained free, with Seed 0, one thread, MIPFocus 1 and SolutionLimit 1. The objective was constant-zero feasibility under the hard original-price cutoff. Original model $T = 6300$ seconds is separate from the experimental window. The call returned normally with status TIME_LIMIT, no integer solution, 258972 final nodes, 160559495 simplex iterations and 318526.526959 work units. Native Runtime was 129600.004 seconds and the charged outer time was 129601.832 seconds. Its zero feasibility bound is not an original-J lower. The full level is therefore UNKNOWN and the table's prior intervals are retained.

After that call closed, an exact rational saved-point check instantiated the fractional-ownership construction at Y3. Its total pickups and drops are both 56; each vehicle owns 1/3 of every changed station and has zero returned load. The same point satisfies every actual row and bound of both admitted matrices, with exact maximum violation zero. It has 102 fractional integer columns, maximum integrality violation 1/3, and original price approximately .23685400930608236. A separate saved readback repeats all exact row and bound checks. Thus the two bare LPs retain this integer stock through fractional ownership even though its integer resource family is excluded. This is no integer resource upper, no limit on stronger integer-valid cuts, and no proof that the global resource minimum strictly exceeds the inventory minimum. It documents a specific obstruction without identifying the unique cause of search cost.

The total scientific outer charge rounds to 37.1 hours, including failed preparation, the normally stopped watchdog/coefficient-repair calls, necessary qualification, and saved-original readers. All receipts are closed; nested Runtime and Work are not charged again. The first exact LP reader failed on an infinite-bound conversion; its original bytes and fee remain, and only the signed infinite-bound readback was corrected. PDF compilation/rendering, pure writing and publication I/O are separate. Current results and exact final costs accompany the saved evidence.

This completes a substantive finite computational negative under the predeclared capacity. Whether $F_R - F_I$ is worthwhile remains open. No new original-objective lower or true route benefit justifies freezing E+R, so the conditional eight-arm panel was not entered. The next research question concerns justified information about integer vehicle ownership or stronger physical travel/load structure. Stable GI-TBC/B remains the default and E+ the enhanced baseline; no production change or complete-arm benefit is claimed.

5 Evidence consumption and the complete method

A whole-level resource lower is a bound for the original inventory scope and can enter the GI-TBC ledger only with that scope and its coverage qualification. A local family proof cannot close all low-objective inventories. A physical route can update U only after the unchanged original-time, station, vehicle, operation and prefix-load checks. A relaxed-time assignment, a log objective or an unfinished callback point does not substitute for either certificate.

Any complete E+R candidate must pay for its own inventory pricing, resource-level computation, route construction and verification. Offline stocks, old assignments and proofs are not free starts for a nominal fresh arm. A uniform rule may use current mathematical bounds and fixed node/iteration capacities; it may not dispatch on instance names, historical wins or a timer switch. A worthwhile structural finding need not become a production mechanism. Stable GI-TBC exposition and the maintained paper continue independently of whether this research branch produces such a candidate.

Attribution and scope. This section continues the support-duration studies (R51–52), the free-fleet/time-oracle studies (R61), the physical-envelope rounded handling repair (R92), full handling with anchors (R103), resource propagation (R150–151), and integer-inventory objective proof consumption (R152–157). Results, scope identities, actual Native matrices, source snapshots, fees and same-agent separate readbacks accompany the present finite study. No reviewer independence, universal performance benefit or default promotion is inferred.